The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Precursor-Generating Substrate Origin Dynamics
Next Benchmark Criterion

Abstract

The current precursor-frontier routing note had fixed the admissible burden order below the theorem forcing the lift-generating substrate precursor package $\mathfrak{P}^{\mathrm{sub,prec}}$ from precursor-generating substrate origin dynamics $\mathfrak{O}^{\mathrm{sub,orig}}$: first try to derive or uniquely force the origin package from still more primitive explicit substrate structure, and if that could not yet be done cleanly, prove a genuinely smaller theorem-level collapse or sharper necessity / uniqueness result strictly inside origin scope. The new origin-forcing theorem has now spent that first admissible route. The precursor-generating substrate origin package $\mathfrak{O}^{\mathrm{sub,orig}}$ is no longer left as an independently inserted stopping point; it is a canonical and unique output of origin-generating substrate ground dynamics $\mathfrak{G}^{\mathrm{sub,grd}}$.

This note records the routing consequence of that new theorem. The next honest benchmark-facing manuscript must therefore go one level below the current origin frontier: first derive or uniquely force the origin-generating substrate ground package $\mathfrak{G}^{\mathrm{sub,grd}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status, or, if that cannot be done cleanly, prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the precursor-generating substrate origin package $\mathfrak{O}^{\mathrm{sub,orig}}$. Another shell-restriction relay, lift-package restatement, precursor-package restatement, split-core restatement, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another origin relay that preserves the same $\mathfrak{O}^{\mathrm{sub,orig}}$ without shrinking or forcing it further does not count next.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language remains paused. The positive result of this note is therefore routing discipline at the new origin frontier.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is no longer at a stage where the precursor-generating substrate origin package $\mathfrak{O}^{\mathrm{sub,orig}}$ should be treated as the default unexplained stopping point. The precursor-forcing theorem had already shown that the lift-generating substrate precursor package $\mathfrak{P}^{\mathrm{sub,prec}}$ is a canonical and unique output of $\mathfrak{O}^{\mathrm{sub,orig}}$. The new origin-forcing theorem has now spent the next admissible upstream move below that frontier: $\mathfrak{O}^{\mathrm{sub,orig}}$ itself is a canonical and unique output of the deeper ground class $\mathfrak{G}^{\mathrm{sub,grd}}$.

The present note records the routing consequence of that theorem. It does not reopen the origin mathematics, and it does not claim a completed first-principles substrate derivation of gravity. It

records only which kind of next manuscript would now count as an honest benchmark-facing gain below the current origin frontier.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the origin frontier.

2. Fixed Inputs

Definition 1 (Origin-frontier fixed inputs)**.** *The criterion recorded here uses the following fixed inputs.*

1. *The benchmark exact-closure package, which fixes the public one-metric GR target of the repository.*
2. *The benchmark gatekeeping layer, including the post-bridge status correction and the upstream-compression safeguard that blocks benchmark-neutral deeper-origin relays from counting as new front-line progress once the benchmark-relevant object is unchanged.*
3. *The preserved precursor-forcing theorem, which already shows that the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$ *is a canonical and unique output of precursor-generating substrate origin dynamics* $\mathfrak{O}^{\mathrm{sub,orig}}$.
4. *The preserved origin-from-ground theorem, which already shows that* $\mathfrak{O}^{\mathrm{sub,orig}}$ *can be recovered canonically from origin-generating substrate ground dynamics* $\mathfrak{G}^{\mathrm{sub,grd}}$.
5. *The new origin necessity / uniqueness theorem, which proves that* $\mathfrak{O}^{\mathrm{sub,orig}}$ *itself is a canonical and unique output of* $\mathfrak{G}^{\mathrm{sub,grd}}$.

Remark 1. *The present note does not replace those manuscripts. It records the routing consequence of reading them together under the active safeguard against recursive depth drift.*

3. Why the Origin Theorem Is the Frontier

Definition 2 (Benchmark-neutral origin relay)**.** *A* benchmark-neutral origin relay *is a manuscript that preserves the same benchmark one-metric output, the same ordinary matter route, the same precursor-generating origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*, the same lift-generating precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*, the same shell-generating lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, the same shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, the same split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, the same selector-law consequence* $\mathfrak{S}^{\mathrm{sel},0}$*, the same combined-response*

consequence $\mathfrak{R}^{\mathrm{comb}}$*, the same ambient-package consequence* $\mathfrak{A}^{\mathrm{bal}}$*, the same trace-core consequence* $\mathfrak{C}^{\mathrm{tr}}$*, the same completion-core consequence* $\mathfrak{C}^{\mathrm{zr,aff}}$*, the same exact-shell affine nucleus consequence* $\mathfrak{N}^{\mathrm{aff}}$*, and the same claim boundary while merely restating* $\mathfrak{O}^{\mathrm{sub,orig}}$*,* $\mathfrak{P}^{\mathrm{sub,prec}}$*,* $\mathfrak{L}^{\mathrm{sub,lift}}$*,* $\mathfrak{S}^{\mathrm{sub,sh}}$*,* $\mathfrak{C}^{\mathrm{sel,split}}$*,* $\mathfrak{S}^{\mathrm{sel,0}}$*,* $\mathfrak{R}^{\mathrm{comb}}$*,* $\mathfrak{A}^{\mathrm{bal}}$*,* $\mathfrak{C}^{\mathrm{tr}}$*,* $\mathfrak{C}^{\mathrm{zr,aff}}$*,* $\mathfrak{N}^{\mathrm{aff}}$*, or the same ground consequence without shrinking or forcing the origin package further.*

Proposition 1 (The origin-forcing theorem is the live frontier)**.** *At the current stage of the primitive-reservoir observer-reduction line, the theorem forcing the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$ *from origin-generating substrate ground dynamics* $\mathfrak{G}^{\mathrm{sub,grd}}$ *is the live benchmark-facing frontier.*

Proof. That theorem changes benchmark-facing status at current scope because it removes the origin package $\mathfrak{O}^{\mathrm{sub,orig}}$ itself as the live internal stopping point. The current bridge datum is no longer merely a deeper package whose precursor consequence was already known; it is now forced from the ground layer. By contrast, a manuscript that preserves the same benchmark package, the same origin consequence, and the same precursor/lift/shell-restriction consequences while merely restating the origin package or another already-forced larger or smaller bridge object does not alter benchmark-facing status unless it adds a comparably sharp gain. Therefore the live frontier is the origin-forcing theorem rather than the earlier precursor-forcing theorem or a later same-origin relay. □

Corollary 1 (Status of the origin package as a next-step target)**.** *Under the current routing safeguard, the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$ *is no longer itself the default next benchmark-facing target.*

Proof. The origin-forcing theorem already proves that $\mathfrak{O}^{\mathrm{sub,orig}}$ is a canonical and unique output of $\mathfrak{G}^{\mathrm{sub,grd}}$. Once that is on record, another manuscript that spends the move on the origin package again does not remove any further benchmark-relevant freedom. The open burden has therefore moved below origin scope. □

4. The Next Benchmark Criterion

Definition 3 (Admissible next benchmark-facing origin manuscript)**.** *At the current origin frontier, a manuscript counts as an admissible next benchmark-facing move only if it respects the following burden order:*

1. *its primary attempt is to derive or uniquely force the origin-generating substrate ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$ *from still more primitive explicit substrate structure in a way that does more than merely relocate the unexplained layer deeper;*

2. *if that cannot be done cleanly at current scope, its admissible fallback gain is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*;*

3. *in either case it preserves the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary, and it introduces neither a second operative metric nor an enlarged low-energy matter law and licenses no stronger first-principles promotion language.*

Theorem 1 (Next honest benchmark-facing task at origin scope)**.** *The next honest benchmark-facing manuscript at the current origin frontier should therefore go one level below that frontier: first try to derive or uniquely force the origin-generating substrate ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$ *from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the precursor-generating substrate origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*. A manuscript that only restates the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*, only restates the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, only restates the shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, only restates the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, only restates the selector law* $\mathfrak{S}^{\mathrm{sel},0}$*, only restates the combined response law* $\mathfrak{R}^{\mathrm{comb}}$*, only restates the ambient package* $\mathfrak{A}^{\mathrm{bal}}$*, only restates the trace core* $\mathfrak{C}^{\mathrm{tr}}$*, only restates the completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*, only restates the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*, or only repeats the same origin consequence while preserving* $\mathfrak{O}^{\mathrm{sub,orig}}$ *without shrinking or forcing it further does not satisfy the criterion.*

Proof. By Proposition 1, the live frontier already lies at the theorem that forces the origin package from ground dynamics. Therefore a second manuscript below it must improve on that status rather than merely accompany it. A deeper-origin manuscript can do so only if it changes benchmark-facing status by more than depth alone, and the first clean way to do that is to force $\mathfrak{G}^{\mathrm{sub,grd}}$ itself from still more primitive explicit substrate structure. If that first option is not yet available cleanly, the routing safeguard allows the fallback of a smaller internal collapse, necessity result, or uniqueness result inside $\mathfrak{O}^{\mathrm{sub,orig}}$. If a manuscript simply reproduces the same origin consequence through another relay while preserving the same origin package without shrinking or forcing it further, the routing rule classifies it as benchmark neutral. □

5. What the Next Move Should Not Be

Proposition 2 (Screened next moves)**.** *At the current origin frontier, none of the following is the default next benchmark-facing move:*

1. *another restatement of the lift-generating substrate precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*;*
2. *another restatement of the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*;*
3. *another restatement of the primitive substrate shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*;*
4. *another restatement of the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*;*
5. *another restatement of the full zero-hidden selector law* $\mathfrak{S}^{\mathrm{sel},0}$*;*
6. *another restatement of the combined response substrate law* $\mathfrak{R}^{\mathrm{comb}}$*;*
7. *another restatement of the completion-balance ambient dynamics package* $\mathfrak{A}^{\mathrm{bal}}$*;*
8. *another restatement of the completion-state shell-trace core* $\mathfrak{C}^{\mathrm{tr}}$*;*
9. *another restatement of the full completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*;*
10. *another restatement of the exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*;*
11. *another origin relay that preserves the same* $\mathfrak{O}^{\mathrm{sub,orig}}$ *without shrinking or forcing it further;*

12. *a manuscript that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger first-principles promotion language.*

Proof. Items (1)–(11) fail the preceding criterion because they do not directly address the current ground-facing burden. They revisit either already-forced larger bridge objects, already-forced smaller bridge objects, or the same origin stopping point rather than removing the remaining origin freedom or proving a still smaller internal datum inside $\mathfrak{O}^{\mathrm{sub,orig}}$. Item (12) violates the fixed claim boundary of the benchmark package and therefore cannot count as a disciplined next step on the active line. The benchmark package remains the exact one-metric GR package already fixed by adoption, so any admissible next manuscript must preserve that boundary while keeping stronger first-principles promotion language paused. □

Remark 2. *This screening statement is not a denial that the precursor-forcing theorem, the lift-forcing theorem, the shell-restriction forcing theorem, the selector-law collapse theorem, the selector-law forcing theorem, the combined-response theorem, the ambient-package theorem, the trace-core theorem, or the completion-core theorem matter historically. It is a routing statement: they are not the honest way to spend the next benchmark-facing move from the current origin frontier.*

6. Conclusion

The positive result of this note is routing discipline at the current origin frontier. The origin-forcing theorem remains the live benchmark-facing gain because it already removes the precursor-generating substrate origin package as the internal stopping point on the active line. The origin package therefore no longer sets the honest next burden, and neither do the already-forced precursor package, lift package, shell-restriction package, split core, selector law, combined response law, ambient package, trace core, completion-core presentation, or exact-shell affine nucleus.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger first-principles promotion language yet.

The next open burden is therefore clear. The next benchmark-facing manuscript should go one level below the current origin frontier: first try to derive or uniquely force the origin-generating substrate ground package $\mathfrak{G}^{\mathrm{sub,grd}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the precursor-generating substrate origin package $\mathfrak{O}^{\mathrm{sub,orig}}$. Another precursor relay, lift relay, shell-restriction relay, split-core restatement, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another origin relay that preserves the same origin package without shrinking or forcing it further is not the clean next manuscript target at current scope.